

AMELOGENESIS TUTORIAL

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TASK: IDENTIFY, STUDY AND PASTE

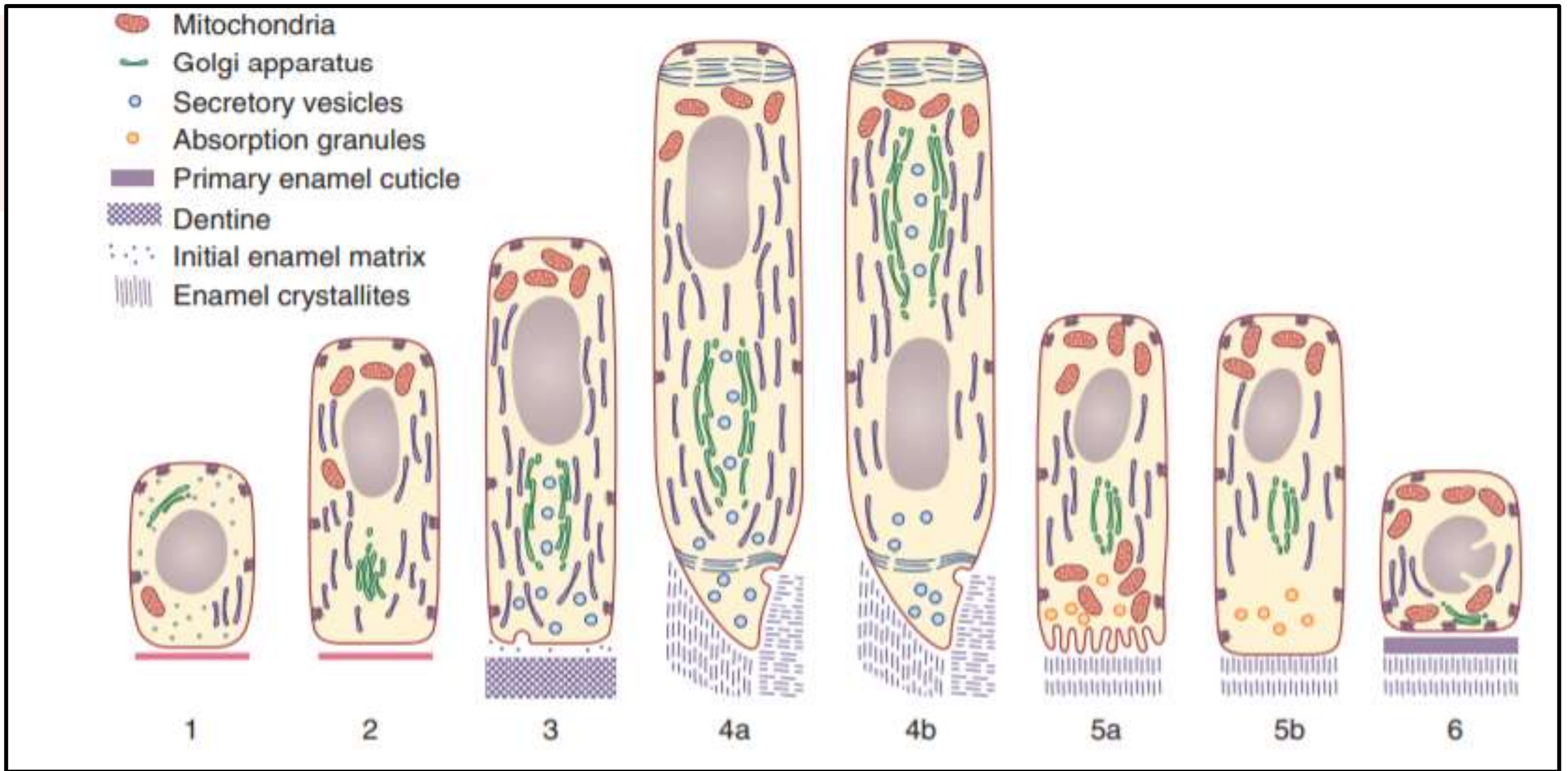


Fig. 22.1 The life cycle of an ameloblast.

Fig. 22.1 Explanation:

- The cells of the internal enamel epithelium (1) start to differentiate, beginning at the future enamel–dentine junction of the cusp tip. The differentiating cell (2) is characterised by a reversed polarity; the cell becomes columnar and the nucleus moves to that part of the cell furthest from the dentine. Secreting organelles form, and the end of the cell adjacent to the dentine becomes the site of secretion. At the
- Next stage (3), the cell secretes the initial enamel component of the enamel–dentine junction. This thin layer will be continuous with the inter-rod enamel of the later-formed tissue. As the cell retreats, the secreting pole becomes morphologically distinct as a pyramidal Tomes process (4a). Crystallites are formed at both surfaces of the process. The proximal region between two processes, deep in the junctional regions, always secretes ahead of the more distal region so that pits surrounded by inter-rod enamel are formed. These are then filled, giving the prism configuration to the tissue. Simultaneous secretion of both organic material and mineral continues until the full thickness of the tissue is formed.
- In this secreting phase, two appearances of ameloblasts can be distinguished by the position of the nuclei within the cell: high (4a) and low (4b). At the beginning of secretion, half the cells are in each form. Towards the end of secretion, most of the high nuclei have moved to a low position, effectively increasing the areas of the ameloblast cells as the surface of forming enamel increases.
- When the full thickness of enamel has formed, ameloblasts lose the secretory extension, the Tomes process (5a). Up to 50% of them die and are phagocytosed by others in the layer.
- The maturation phase lasts two to three times longer than the secretory phase. There is a regular, repetitive modulation of cell morphology between a ruffled (5a) and a smooth (5b) surface opposed to the enamel.
- Once the maturation changes are complete, the cells regress in height (6). At this stage, they serve to protect the enamel surface during eruption and later will contribute to form the junctional epithelium.

Fig. 22.12 The relationship between Tomes processes and enamel prism formation

Explanation:

The enamel of the core of the prism and the prism boundary/inter-rod regions differ largely in the orientation of crystallites. This is determined by the shape of the Tomes process. The 'prongs' of Tomes process are two distinct ends. The distal end of Tomes process forms the core of the prism (rod) and the proximal end makes the boundary of prism (inter-rod region).

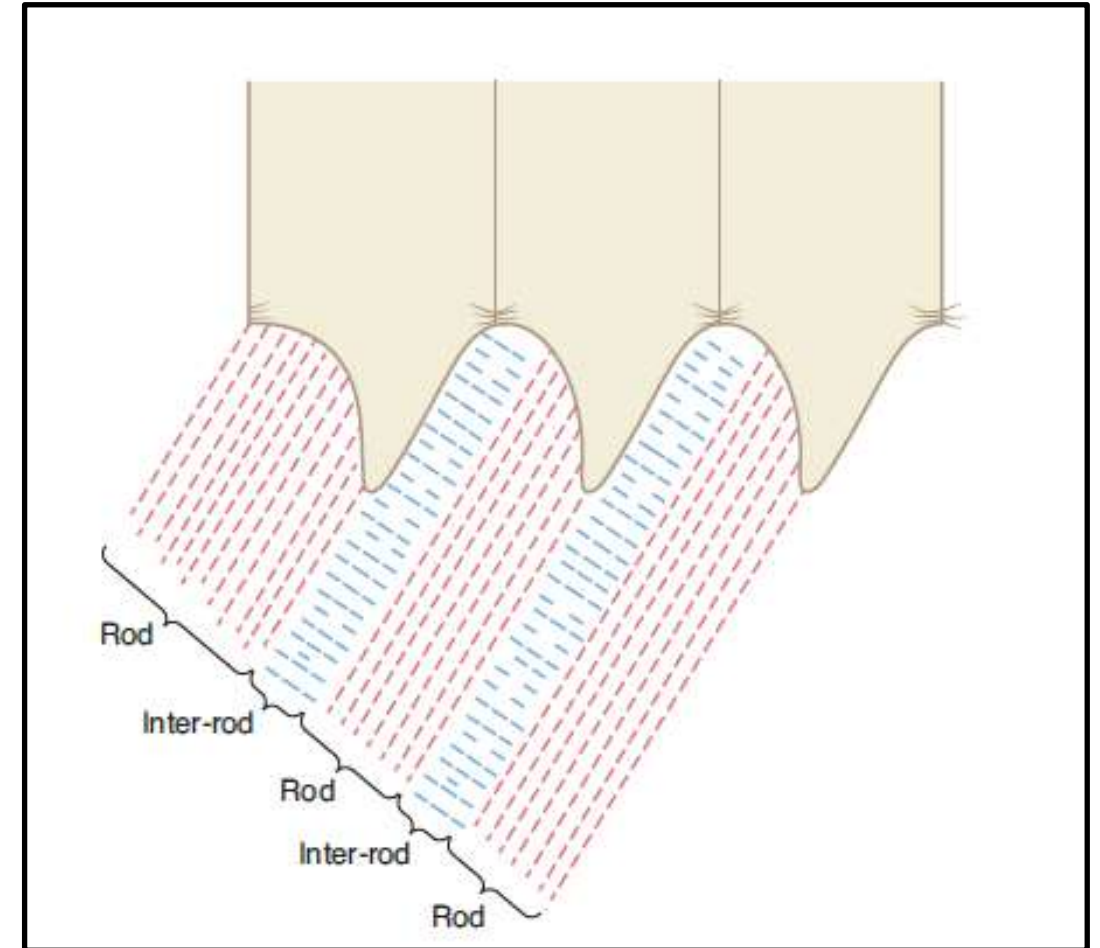


Fig. 22.9 Demineralised section of developing enamel showing the cone-shaped Tomes process at the distal end of each ameloblast



- A- Ameloblast
- B- Enamel Matrix
- C- Dentine
- D- Stratum Intermedium
- E- Terminal bar apparatus

Explanation: The arrowhead shows Tome's process giving the mineralizing front of enamel a picket-fence appearance.

IDENTIFY THE FOLLOWING CLINICAL CONSIDERATIONS



Amelogenesis Imperfecta

1. Hypoplastic AI:

Defect in enamel matrix formation

Enamel is thin but relatively hard

Pits, grooves, or reduced enamel thickness

2. Hypomaturational AI

Defect in enamel maturation

Enamel thickness is normal but softer than normal

Mottled white-yellow-brown appearance

Enamel may chip from dentin

3. Hypocalcified AI

Defect in initial mineralization

Enamel thickness is normal at eruption but very soft

Rapid wear and staining

IDENTIFY THE FOLLOWING CLINICAL CONSIDERATIONS



Tetracycline staining is an intrinsic discoloration of teeth caused by the incorporation of tetracycline antibiotics into developing enamel and dentin during tooth formation. This results in a band of brown enamel (pigmentation).

IDENTIFY THE FOLLOWING CLINICAL CONSIDERATIONS

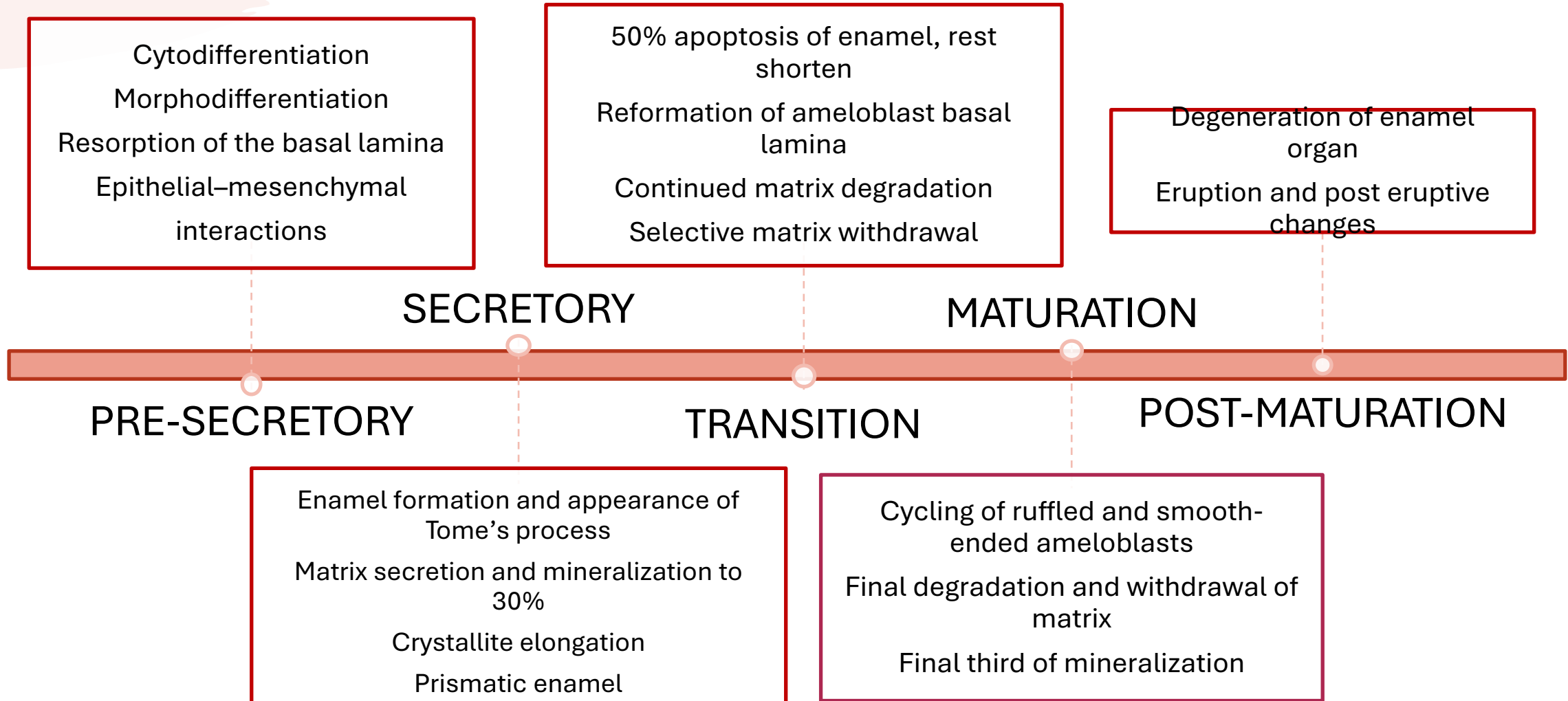


Flourosis occurs when there is ingestion of flouride levels above 5 parts per million and is characterised by the presence of a diffuse white opacity. A high level of dietary fluoride has resulted in much of the enamel becoming opaque in patches, giving a 'mottled' appearance.



REFERENCE/STUDY MATERIAL

STAGES OF AMELOGENESIS



Reciprocal Induction

Inner enamel epithelial cells at late bell stage differentiate → tall columnar preameloblasts → signaling molecules including (TGF- β) → dental papilla ectomesenchymal cells differentiate into odontoblasts → predentin is formed → mineralization of dentin → first dentin mineral crystals → induces differentiation into secretory ameloblasts → enamel formation.

ENAMEL PROTEINS

Protein	Percentage / Abundance	Key Features	Main Functions	Stage of Action
Amelogenin	~90% of enamel matrix	Hydrophobic; rich in proline, leucine, histidine; secreted as 25 kDa	Controls crystal length, orientation; prevents premature thickening	Secretory → early maturation
Enamelin	~3–5%	Largest enamel protein; acidic.	Crystal nucleation; early mineralization; acts near EDJ & prism core	Early secretory
Ameloblastin	Second most abundant	Secreted with amelogenin; rapidly degraded	Maintains ameloblast polarity; forms prism sheath	Secretory stage
Amelotin	Small amount	Expressed during maturation; non-structural	Hardens final enamel surface	Maturation
Tuftelin	Minor	Acidic; rich in glutamic & aspartic acid; early protein	Signalling molecule; epithelial–mesenchymal interaction	Presecretory / early secretory

PROTEOLYTIC ENZYMES

Enzyme	Stage Active	Function	Role in Amelogenesis
MMP-20 (Enamelysin)	Secretory → early maturation	Partial degradation of enamel proteins	Produces functional protein fragments for controlled crystal growth
KLK-4 (Kallikrein-4)	Maturation	Complete degradation of remaining proteins	Clears organic matrix → allows crystal thickening